Mechanical Analysis of Sustainably Derived Resins & Individual Natural Fibers for SUstainable Manufacturing of AirCraft (SUMAC) Composites

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Advanced Materials and Processing Branch
NASA Langley Research Center

Challenge

- Aircraft industry uses tons of carbon composites
  - Commonly made with thermoset resins
    - Not recyclable or repairable
    - Discarded once expended
  - Negative environmental impact
    - Poor and slow biodegradability
    - Releases pollution when degrading
- Aircraft engine and fuel efficiency have been optimized for sustainability
  - Limited research has been done for recycling the aircraft
- Sustainable aircraft composites is a new and growing field
  - Little precedent or past research compared to other common composites (carbon fibers, glass fibers, and epoxies)

We only have one home, for now, lets take care of it.

STRUCTURES AND MATERIALS

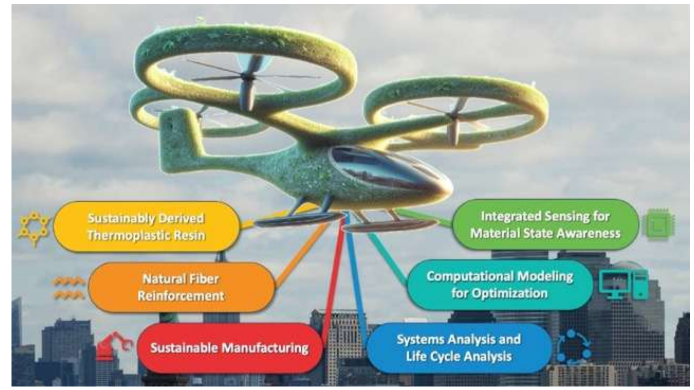
Objective

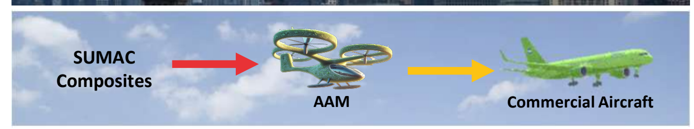
❖SUstainable Manufacturing of Aircraft (SUMAC)'s objectives:
- ❖ Develop sustainably derived thermoplastic composite technologies for future aviation
  - ❖ Advanced air mobility (AAM) vehicles
  - ❖ Advance the sustainable manufacturing of aircraft in six key focus areas

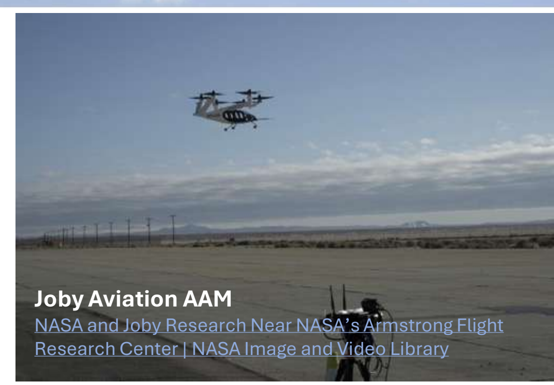
❖Enoch's objective:
- ❖ Characterize the mechanical properties of various natural fibers and thermoplastic resins
  - ❖ EnviroTextiles Hemp Fibers
  - ❖ BComp Flax Fibers & A&P Braided Flax Fibers
  - ❖ Mango Materials Polyhydroxy Butyrate (PHB)
  - ❖ Pond Polylatic Acid (PLA)
  - ❖ Arkema Rilsan/Polyamide 11 (PA-11)
- ❖ Assist sensing and modeling team at Ames Research Center (ARC) to create accurate computation natural composite models

❖What this means for SUMAC:
- ❖ Set a precedent for natural composite properties and researching methods
- ❖ Identify the best materials for sustainable composites

SUMAC's Objectives:







Joby Aviation AAM
[NASA and Joby Research Near NASA's Armstrong Flight Research Center | NASA Image and Video Library]

Approach

- Mechanical Testing on Natural Fibers and Biodegradable Thermoplastic Resins
  - Develop procedure for tensile testing individual natural fibers
    - Crucial for characterization of the strongest natural fibers
    - Procedure is transferable to non natural fibers as well
    - The experimental data are fed into computational models to improve simulation capabilities and reduce time and cost
- Image Segmentation and Analysis
  - Additional characterization of natural materials

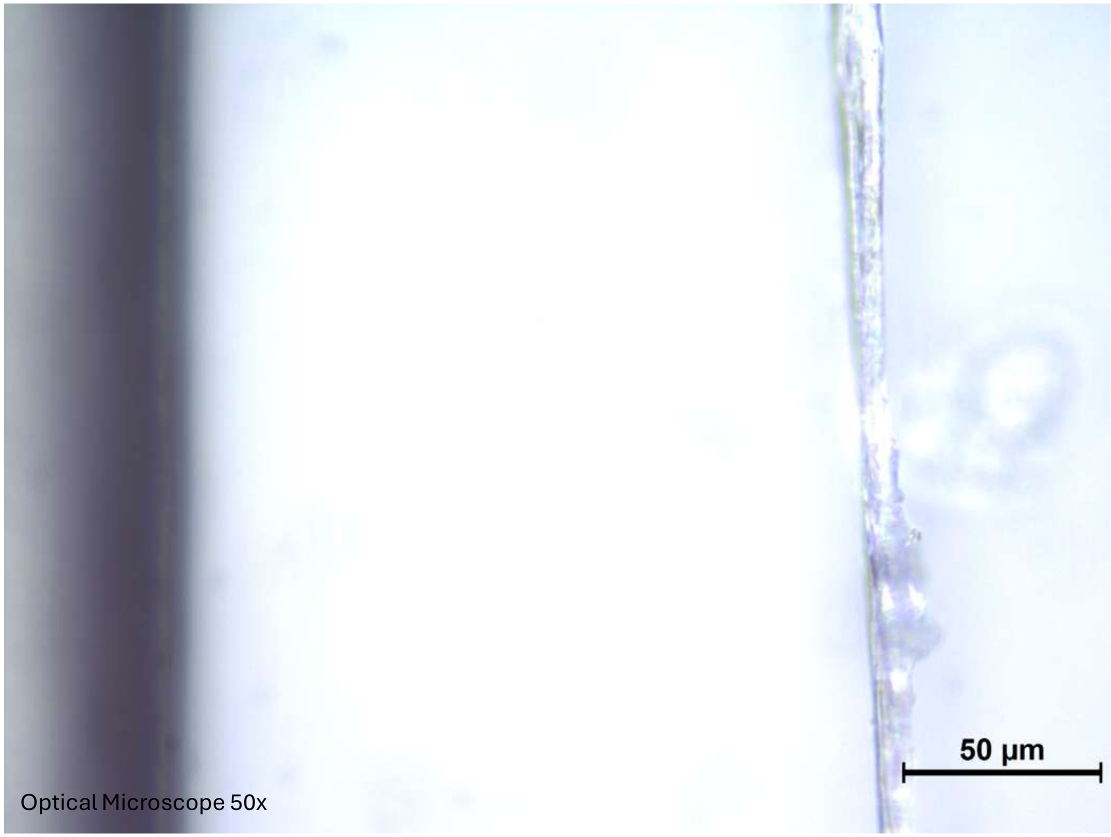
STRUCTURES AND MATERIALS

Mechanical Testing of Individual Natural Fibers

A Closer Look at Natural Fibers

On the right is one of SUMAC's individual natural fibers, specifically it's semi bleached Hemp B608 from the company EnviroTextiles.

Key characteristics to notice are its inconsistent shape and diameter.



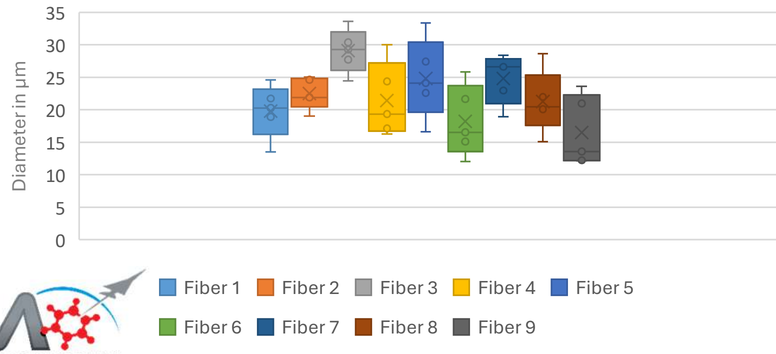
Optical Microscope 50x

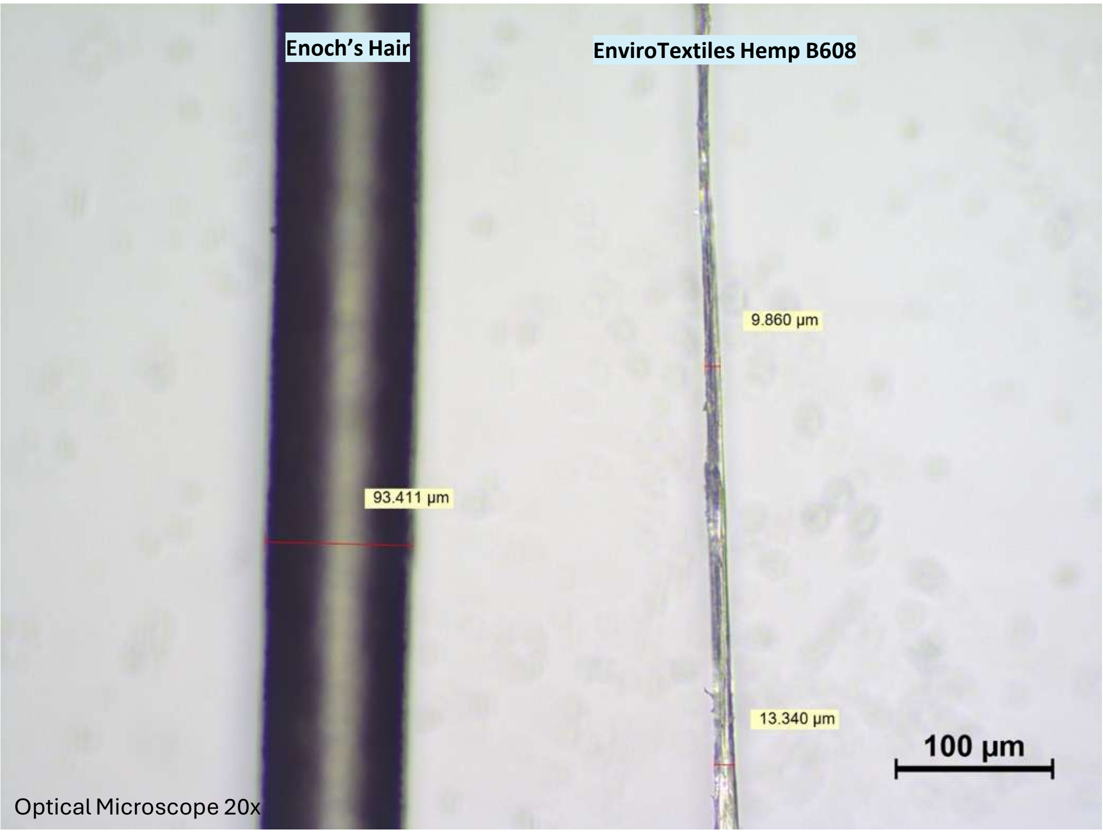
A Closer Look at Natural Fibers

And the black object on the left is my hair… Individual natural fibers are extremely small. Ranging from 10-30 μm.

In addition to being small, they vary greatly in diameter making them a challenge to characterize.

HEMP 1 INDIVIDUAL FIBER DIAMETER FROM OPTICAL MICROSCOPE



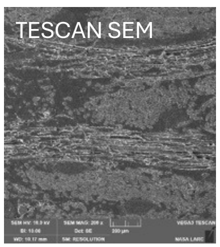


STRUCTURES AND MATERIALS

A Closer Look at Natural Fibers

Composites

Optical Microscope 5x



SUMAC composite
A&P Flax Rilsan

Fabrics

Hemp B601

Optical Microscope 5x

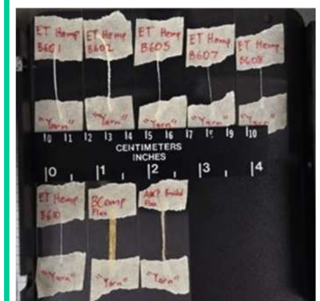
Hemp B602

Optical Microscope 5x

BComp Flax

Stereo Microscope

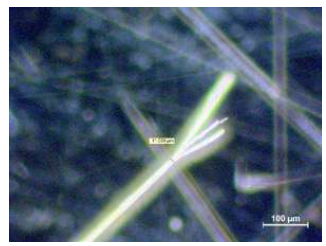
Yarns



Hemp B601

Individual Fibers

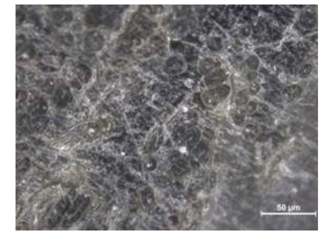
BComp Flax, Stereo Microscope



BComp Flax, Optical Microscope 20x

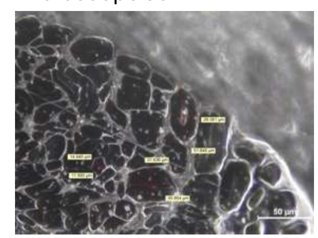
Fiber Cross Sections

Hemp B608 Optical Microscope 50x



BComp Flax Optical Microscope 50x

Hemp B610 Optical Microscope 50x



Hemp B605 Optical Microscope 50x

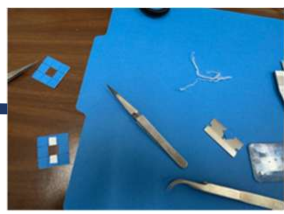
STRUCTURES AND MATERIALS

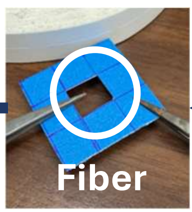
Individual Natural Fiber Tensile Testing Procedure

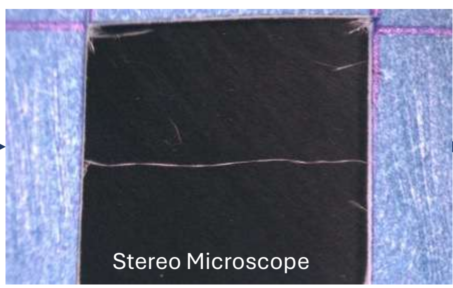
Some of the challenges I faced which warranted creating this procedure were:

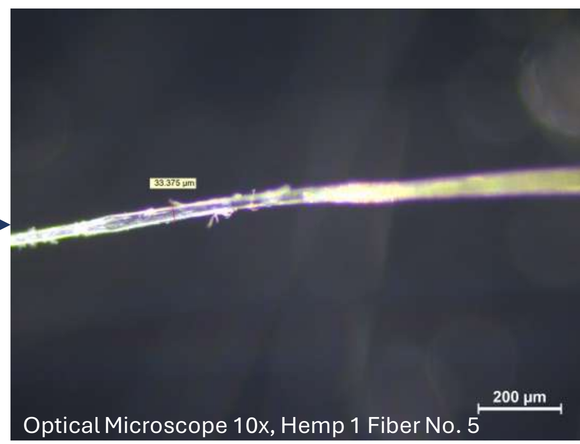
- ❖ Handling such small fibers
- ❖ Separating bundles & yarn into individual fibers
- ❖ Gripping individual fibers for tensile testing

Individual Fiber Separation and Preparation for Tensile Testing

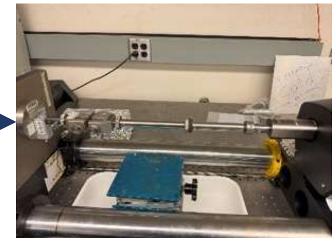






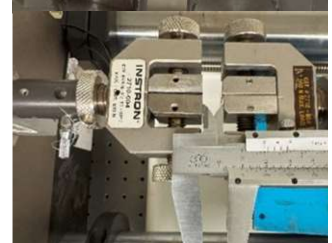


Optical Microscope 10x, Hemp 1 Fiber No. 5



Instron 5848 with 5N load Cell and extension arms used for fragile sample, natural fibers.

Coupon/paper with individual fiber cut right before tensile testing to put load on the fiber.



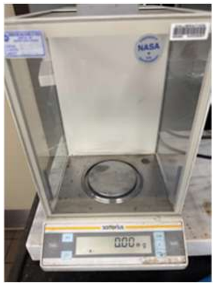
Gauge length measured after preload is applied to the fiber, for most accurate ramp rate, gauge length, and eliminating slack.

Finally, the most crucial part, measure the diameter where fracture occurred. This gives an accurate cross-sectional area for the strength calculation. This is necessary since natural fiber diameters vary so much.

Natural Fiber Mechanical Properties

I charactered many individual fibers and yarn

- ❖ Identified the most favorable mechanical characteristics
  - ❖ Tensile Strength, % Elongation at Break, Modulus, Force/Linear Density, and Toughness



Scale used to calculate linear density.

Individual Fiber Sample & Notes	Tensile Strength (MPa)	% Elongation at Break	Young's Modulus (GPa)
ET Hemp B608	800.4 ± 366.4	3.0 ± 0.4	35.1 ± 1.7
BComp Flax	880.2 ± 596.3	2.4 ± 0.8	42.1 ± 21.7
Hemp 1(Bleached)	518.1 ± 318.3	3.0 ± 0.4	20.7 ± 11.3
A&P Braided Flax	906 ± 768.5	2.0 ± 1.0	52.9 ± 27.2
ET Hemp B607	492.6 ± 233.6	4.4 ± 1.5	15.8 ± 4.3
ET 100% Hemp twill	1238.5 ± 719.1	3.2 ± 0.8	47.5 ± 22.1
Literature Hemp	~500-900	~1.2-3.5	~60
Literature Flax	~340-1500	~1.2-3.2	~30-80

$$tex = \frac{grams}{km}$$

Note:
Yarn's Tensile Strength is only a ballpark value. Measuring diameters for yarn is highly subjective. Force/linear density (tex) is better for yarns.

Yarn Sample & Notes	Force/LinearDensity (N/tex) [N/(kg/m)]	% Elongation at Break	Young's Modulus (GPa)	Linear Density (tex) 1 tex = 1 gram/km	Tensile Strength (MPa) Diameter is subjective, approximate
BComp Flax Yarn	0.172 ± 0.012 [172238 ± 19657]	1.5 ± 0.1	12.0 ± 1.7	273.0 ± 77.4	143.2 ± 23.0
A&P "Braided/Woven" Flax Yarn	0.082 0.031 [82342 ± 30739]	1.4 ± 0.4	4.3 ± 1.4	171.8 ± 91.6	45.5 ± 19.5
"Hemp" 1 Yarn	Not assessed, unknown fiber origin, decided not to proceed	4.5 ± 0.9	7.51 ± 3.3	Not assessed, unknown fiber origin, decided not to proceed	259.7 ± 32.8
"Hemp" 2 Yarn	Not assessed, unknown fiber origin, decided not to proceed	4.9 ± 1.1	7.19 ± 3.6	Not assessed, unknown fiber origin, decided not to proceed	253.8 ± 134.8
"Hemp" 3 Yarn	Not assessed, unknown fiber origin, decided not to proceed	13.1 ± 8.3	5.08 ± 2.5	Not assessed, unknown fiber origin, decided not to proceed	228.8 ± 80.7
EnviroTexiles Hemp B601 Yarn	0.215 ± 0.021 [214686 ± 20611]	6.1 ± 0.7	7.1 ± 1.3	148.0 ± 23.4	329.0 ± 54.5
EnviroTexiles Hemp B602 Yarn	0.071 ± 0.012 [70668 ± 11918]	10.1 ± 1.6	1.0. ± 0.2	267.0 ± 13.5	73.1 ± 15.8
EnviroTexiles Hemp B605 Yarn	0.207 ± 0.050 [207062 ± 49517]	7.5 ± 1.2	3.6 ± 1.0	82.8 ± 46.3	198.9 ± 33.1
EnviroTexiles Hemp B607 Yarn	0.222 ± 0.023 [222045 ± 23066]	9.25 ± 1.4	5.3 ± 0.9	55.6 ± 11.4	381.5 ± 67.2
EnviroTexiles Hemp B608 Yarn	0.284 ± 0.028 [284286 ± 28314]	6.7 ± 1.2	6.1 ± 1.0	26.0 ± 4.3	343.7 ± 38.3
EnviroTexiles Hemp B610 Yarn	0.230 ± 0.051 [230273 ± 51278]	5.7 ± 1.3	8.7 ± 3.6	24.5 ± 7.3	310.7 ± 103.7
EnviroTexiles 100% Hemp Yarn	0.295 ± 0.340 [295023 ± 33508]	4.1 ± 0.8	10.3 ± 2.1	32.4 ± 13.2	378.0 ± 101.4

STRUCTURES AND MATERIALS

Natural Fiber Mechanical Properties

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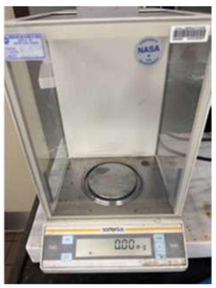
  - Tensile Strength, % Elongation at Break, Modulus, Force/Linear Density, and Toughness

Number of Samples Tensile Tested

128 Individual Fibers, 134 Yarns

Totaling:

262 samples



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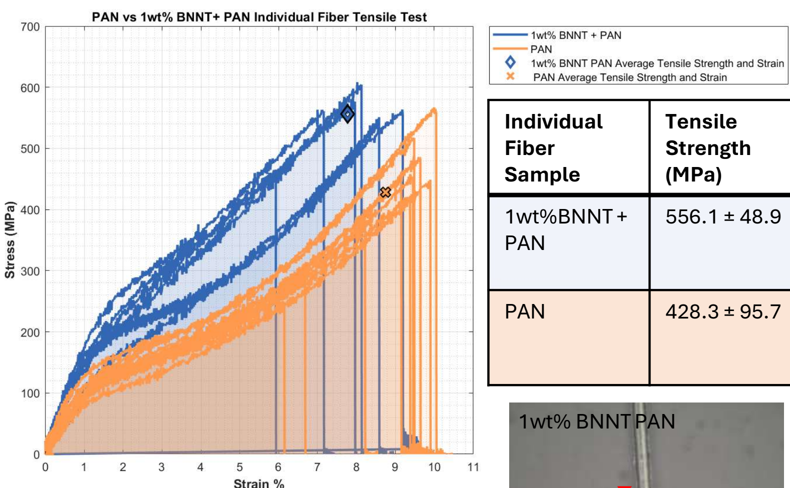
STRUCTURES AND MATERIALS

Gaining a Reputation

Through working with SUMAC, I gained a reputation for my technical and procedural capabilities to test individual fibers. Assisted other projects including:

- ❖ Boron Nitride Nanotube Polyacrylonitrile (PAN) Nanocomposite research
- ❖ Dragonfly Ablative Carbon Phenolic hybrid yarn & individual fiber for Saturn's moon, Titan

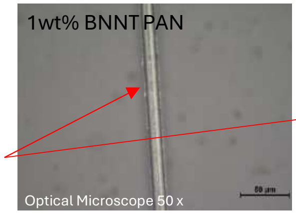
The data from other projects also validated my procedure.

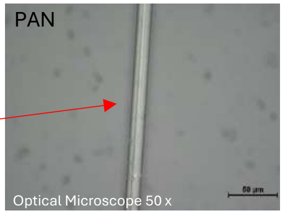


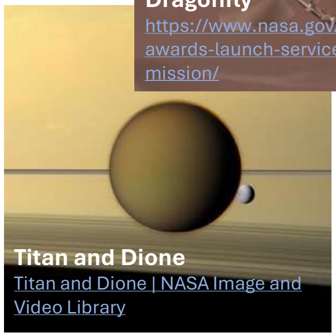
Individual Fiber Sample	Tensile Strength (MPa)	% Elongation at Break	Young's Modulus (GPa)	Toughness (MPa)
1wt%BNNT + PAN	556.1 ± 48.9	7.8 ± 1.2	12.5 ± 2.4	248279.5 ± 40404.1
PAN	428.3 ± 95.7	8.8 ± 1.4	7.1 ± 1.3	201274.9 ± 58573.8

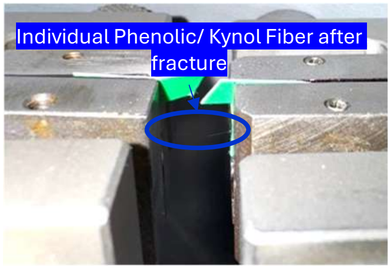
STRUCTURES AND MATERIALS

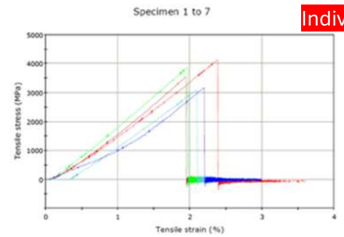
BNNT PAN seems spikier than the PAN, the edges are more rough

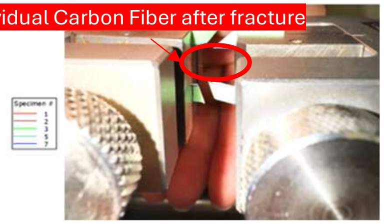




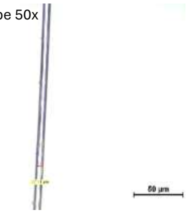








Optical Microscope 50x

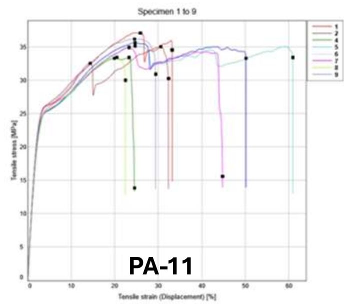


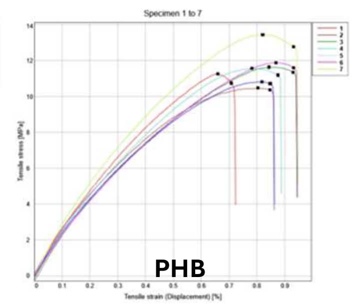
Individual Fiber Sample	Tensile Strength (MPa)	% Elongation at Break	Young's Modulus (GPa)
Carbon Fiber	3568.2 ± 446.7	2.1 ± 0.2	205.2 ± 14.0

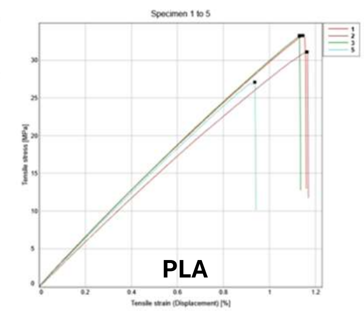
Sustainably Derived Resins Mechanical Properties

Tensile tested sustainably derived resins. Dried samples and non dried samples.

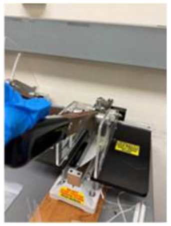
- ❖ Dried PA-11 had significant lose in elongation at break, while a noticeable increase in Tensile Strength and Modulus
- ❖ PHB, PLA difference were within standard deviation- drying seems to have very little effect on them





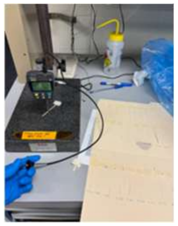


Polymer Preparation for Tensile Testing
Polyamide 11 (PA-11), Polyhydroxy Butyrate (PHB), & Polylatic Acid (PLA)



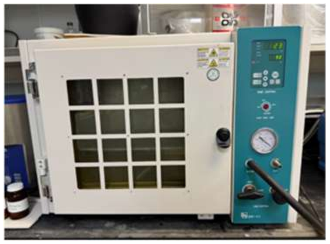
Cut film to 3mm width

Image of cut films, stored at room temperature and humidity.



Film thickness measured, used smallest thickness for tensile test cross section

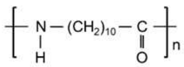
Polymer Drying Process



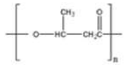
Polymers dried using vacuum chamber set to -0.08 MPa and 25°C for 1 hour.

Polymers tested:

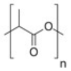
PA-11



PHB



PLA



Sample:	PA-11 @ RT & Humidity >24hr	Dried PA-11 in -0.08MPa vacuum 25°C for 1 hr	PHB @ RT & Humidity >24hr	Dried PHB in -0.08MPa vacuum 25°C for 1 hr	PLA @ RT & Humidity >24hr	Dried PLA in -0.08MPa vacuum 25°C for 1 hr
Tensile Strength in MPa	33.5 ± 2.0	35.1 ± 1.3	14.2 ± 1.6	11.6 ± 1.0	36.2 ± 4.8	31.2 ± 2.9
% Elongation at Break	154.6 ± 60.5	24.7 ± 2.9	1.0 ± 0.1	0.8 ± 0.1	1.3 ± 0.2	1.1 ± 0.1
Modulus in GPa	0.9 ± 0.1	1.2 ± 0.1	2.5 ± 0.1	2.6 ± 0.2	3.0 ± 0.2	3.2 ± 0.1

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Image Segmentation & Analysis

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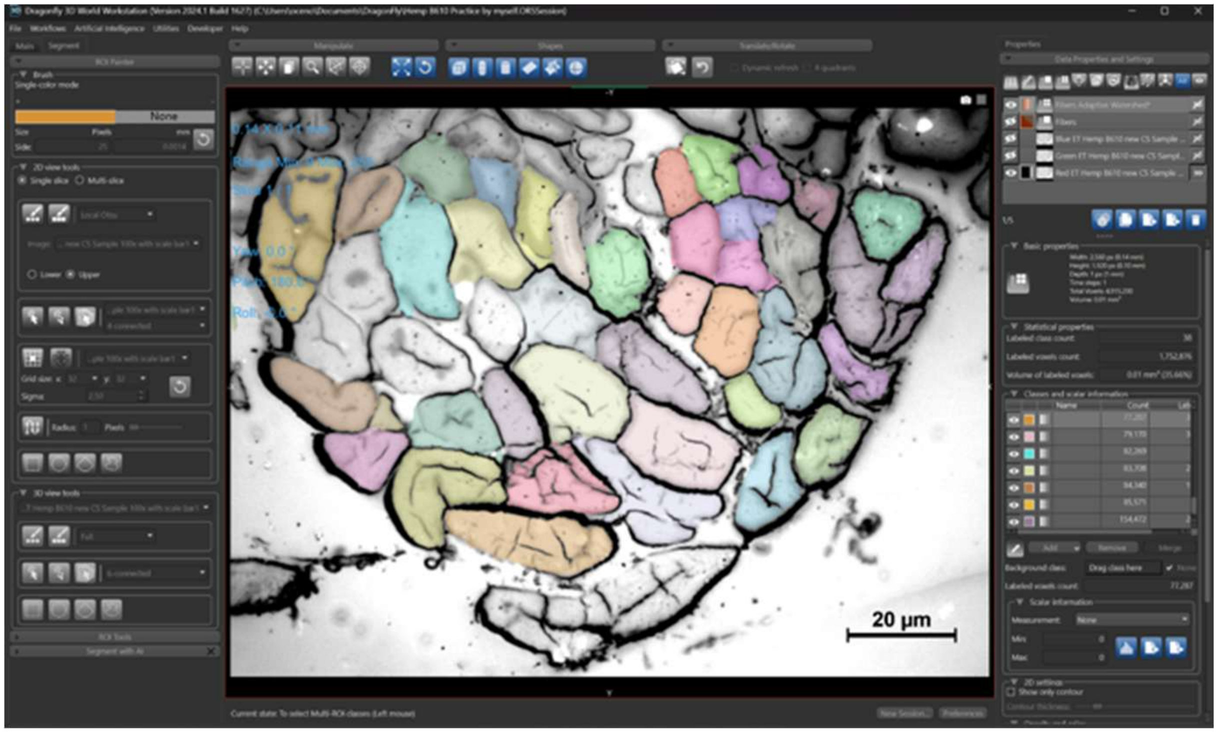
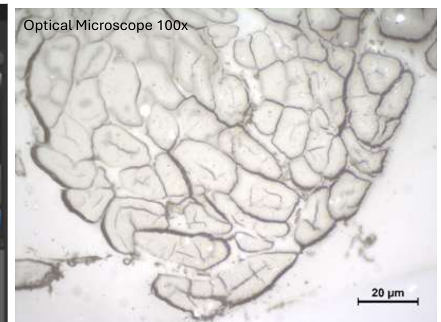
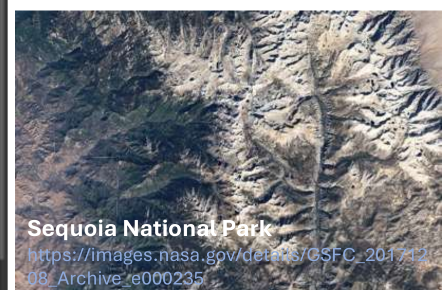
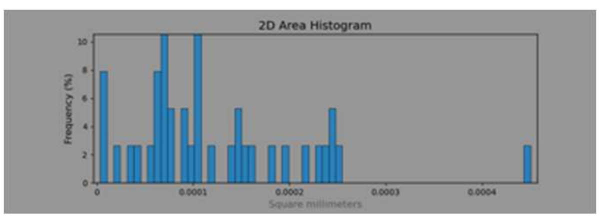
NASA Langley Research Center

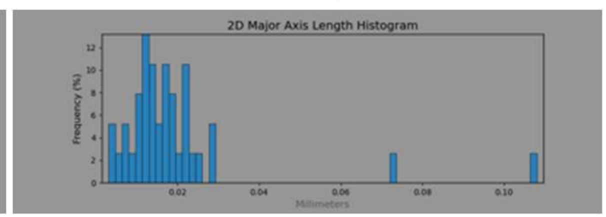
DragonFly Software Image Segmentation











- ❖ Segmentation used to find highly accurate area and diameters of natural fibers
- ❖ DragonFly's Software watershed method to segment fibers, EnviroTextiles Hemp B610
- ❖ Watershed method creates a topographic map of image brightness to segment

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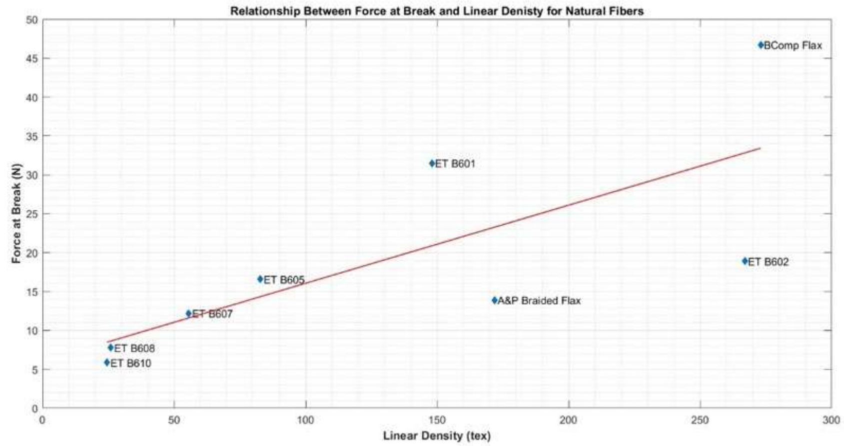
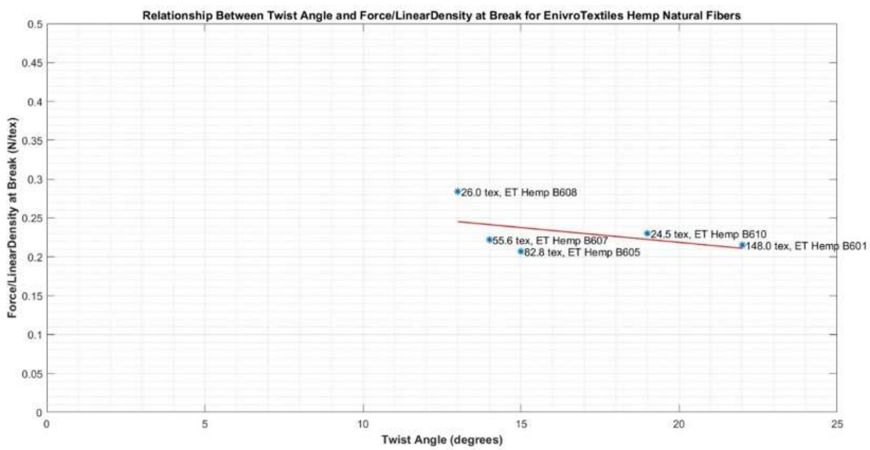
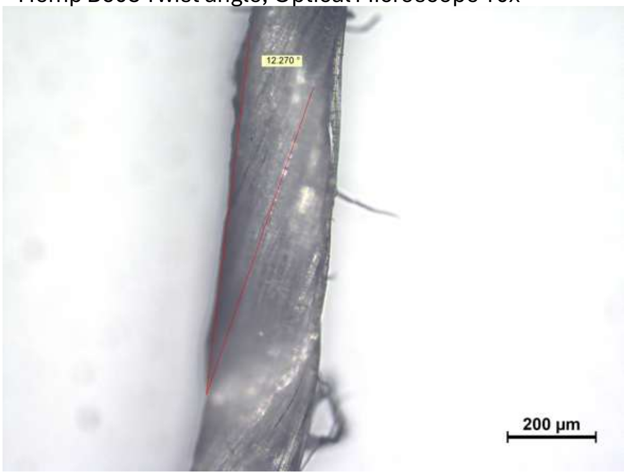
Yarn Twist Angle Mechanical & Image Analysis





- ❖ Yarn force/linear density has negative correlation with twist angle
- ❖ Force at break has positive correlation with with linear density

Hemp B608 Twist angle, Optical Microscope 10x



Results & Analysis

Results

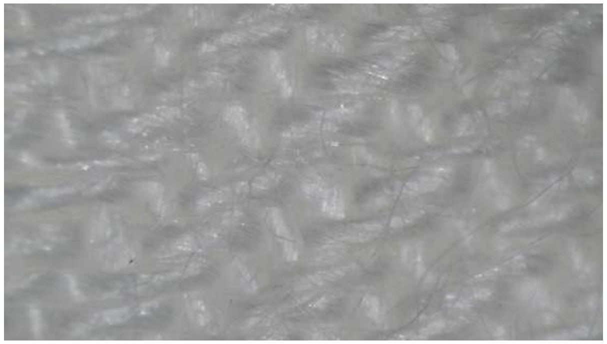
- Natural Fibers have large variations in size, shape, cross sectional area, and most properties (mechanical & thermal)
- On average flax fibers have a higher tensile strength and modulus than hemp, while hemp has a larger % elongation at break
  - This trend is not always true, as seen by EnviroTextiles 100% Hemp Twill, ~1200MPa Tensile Strength
- Polyamide 11 shows promising mechanical properties
  - Similar tensile strength to Polylactic Acid when dried
  - Very ductile, thus absorbs large amounts of energy before fracture (crucial for aircraft materials)
  - Has lowest modulus out of the three resins

Analysis

- Literature reports flax being stronger than hemp, which is has been confirmed from my testing
  - However, EnviroTextiles 100% Hemp Twill is an exception
- Since hemp requires less energy, water, and is space to grow, it is favorable over flax
  - A hemp fiber potentially stronger than flax is an exciting discovery for SUMAC

EnviroTextiles 100% Hemp Twill Stereo Microscope

EnviroTextiles 100% Hemp Twill Stereo Microscope



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Next Steps/Outlook/Future Work

- ❖Continue to gather data on new natural fibers and resins
- ❖Attempt to decrease standard deviation further
  - ❖Utilizing image segmentation cross sectional areas for the most accurate tensile strength
- ❖Segment Anything Microscopy by Meta for streamlined and accurate image segmentation
- ❖Down select the best natural fiber and sustainably derived resin
  - ❖Create a composite with it, and compare with carbon and glass fiber composites
- ❖SUMAC is broadening. Every discovery leads to more questions, there is a big future ahead for SUMAC.

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